UNIVERSITI KEBANGSAAN MALAYSIA

The National University of Malaysia

STQD6014

DATA SCIENCE

PROJECT 1

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PART 1:

INTRODUCTION TO DATA SCIENCE AND ALGORITHMS

Question 1:

Data science is a field of study that focuses on gathering, organizing, analyzing, visualizing, managing, and preserving vast amounts of data. It includes data analysis as a crucial part of the skill set needed for many positions in this field, though it's not the only one. Data scientists play active roles in the design and implementation work of four related areas which are data architecture, data acquisition, data analysis and data archiving.

Data architecture assists the system architect by offering suggestions on how the data should be arranged and routed to facilitate data analysis, visualization, and distribution to the right parties.

Data acquisition is how data is gathered and, more crucially, how it is presented before analysis and presentation. Before the data can be profitably evaluated, it must be represented, transformed, grouped, and linked. Data scientists actively participate in all these processes.

Data analysis involves summarization of the data, using portions of data (samples) to make inferences about the larger context, and visualization of the data by presenting it in tables, graphs and even animations. Even though these activities involve a lot of technical, mathematical, and statistical elements, the final audience for data analysis is usually a person or individual, thus strong communication skills are required.

Data archiving is the task of preserving gathered data in a way that makes it highly reusable. It is challenging since it is very difficult to predict all of the potential uses of the data in the future.

Question 2:

I work as a Department Manager for GreenTech Solutions, a green technology company that specializes in creating eco-friendly and sustainable products to reduce the impact to the environment. Our main product is renewable energy systems (such as solar panels, wind turbines and battery storage systems), eco-friendly consumer products (such as biodegradable cleaning products and sustainable packaging) and smart home solutions (such as smart energy meters, automated lighting systems and smart plugs).

I would like to analyze customer preferences in order to increase customer's loyalty, improve my brand image and ensure satisfaction for clients through effective marketing strategies. To achieve that analysis, the role of data scientist is very useful to investigate and understand the customer preferences and align our products with the purchase demand.

Data scientists will conduct customer preference analysis. This analysis includes product popularity (which products sold are the best-selling), desired features (such as energy efficiency rating of an appliance) and purchase behavior (seasonal trends in product adoption by season or region).

Next, data scientists are going to collect information from various kinds of sources such as sales data (revenue, product categories and transaction information), customer feedback (ratings, reviews and surveys) and external sources (market trends and sentiment on social media). To enable robust insights, data scientists will ensure that the data is clean, consistent and consistent across several sources before proceeding to analysis.

Then, data scientists will proceed with data analysis techniques that consist of segmentation trend analysis, sentiment analysis and predictive modelling. Segmentation is the process of utilizing clustering algorithms to group customers according to their demographics, preferences and purchasing habits. Time series analysis is used in trend analysis to figure out how product preferences changed over time. Sentiment analysis is the technique of monitoring reviews or social media posts to determine customer feedback about a product. Predictive modelling is the process of forecasting which products consumers are likely to enjoy or buy using machine learning algorithms.

Data scientists may establish recommendation systems that suggest products based on the preferences of certain customers.. For example, a customer who is interested in solar panels might also be suggested to use a smart energy meter.

Data scientists may forecast market trends by forecasting future demand for green technologies and direct product development by examining external data such as global energy regulations and climate action plans. The results will be presented to stakeholders using dashboards, graphs and reports. I as the Manager can decide on price, marketing strategies and product development through this approach.

My company will gain advantages from this process in terms of product lines, marketing campaigns and customer support since it will optimize the features and design of the product, assist in the launch of new products and develop marketing strategies to meet the needs of the customer.

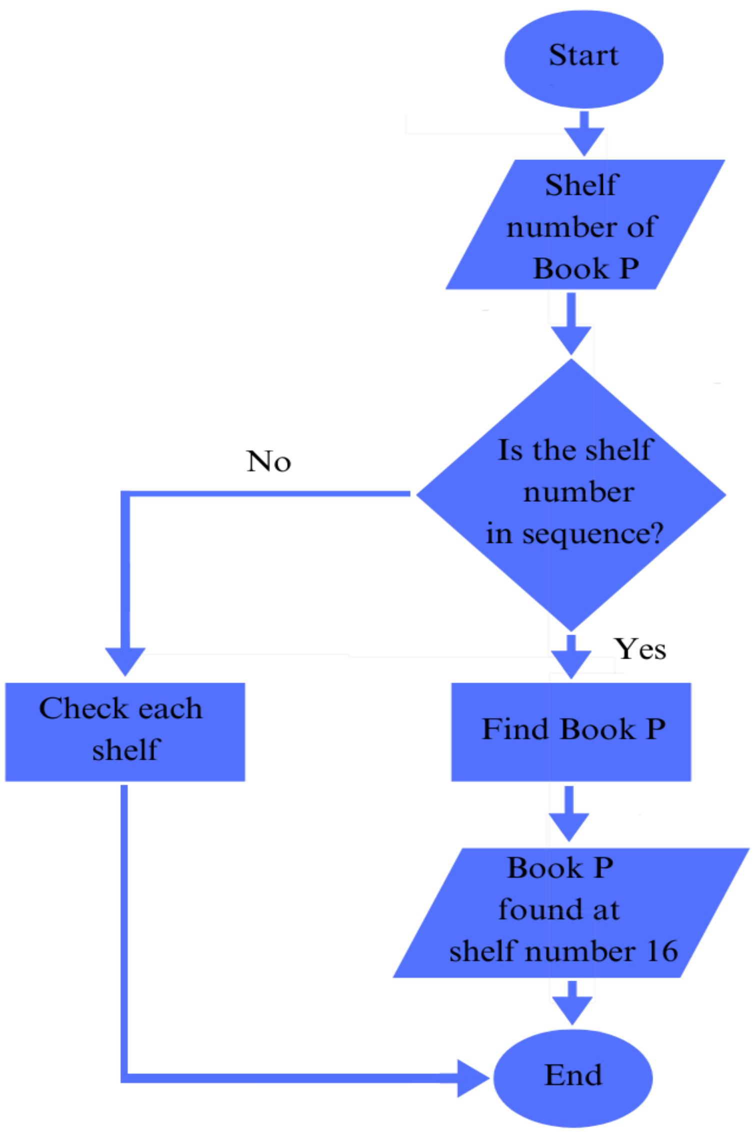
Question 3:

(a).

Problem:	Find the location of Book P in a library that contain 26 shelves
Input:	Shelf number of Book P
Processing:	Decide if the book's location is in sequence order or not
Output:	Book P found at shelf number 16

(b)

Using flowchart:



Using Pseudocode:

Program: Find location of Book P in a library that contain 26 shelves

Find the shelf number of Book P

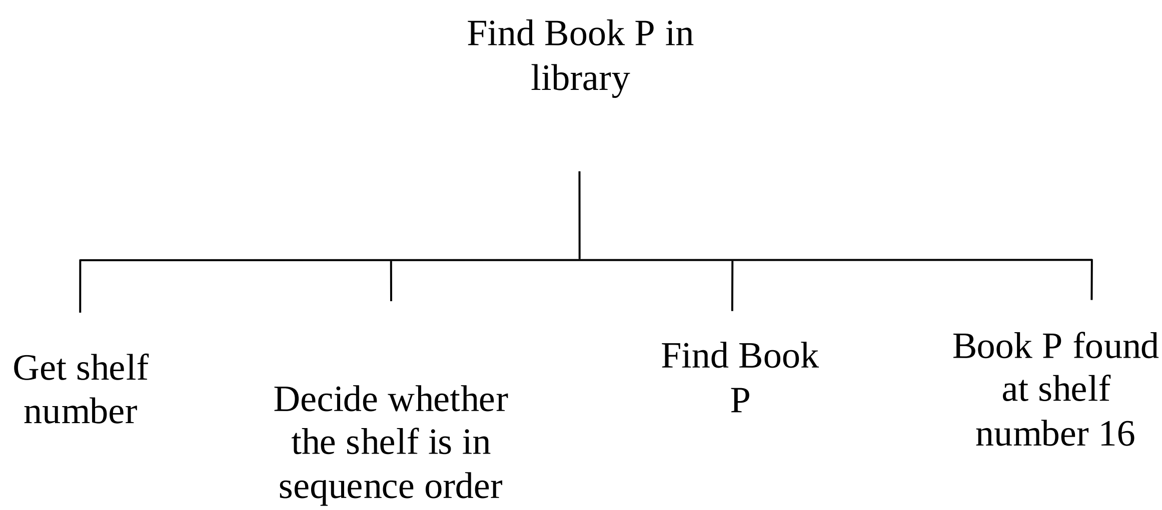
if the book is in sequence order

        find at shelf number 16

else

        check each shelf

Using Hierarchy Chart:



Question 4:

Two examples of current data technology are Amazon Redshift and Power BI. Amazon Redshift is a fully managed cloud data store that is responsible for storing products by using business intelligence and SQL tools. Both tools have a role in analyzing large datasets. Redshift has a high performance inquiry architecture. Its function is to run complex analytical inquiry on petabytes of unstructured and structured data. Its features include data compression, parallel query execution and columnar storage to boost the performance. Redshift is considered a flexible solution for big data analytics and business intelligence due to its interfaces with a number of data sources and analytics tools.

Power BI is a set of software services, applications and connectors that combine to create active, attractive and meaningful insights from different data sources such as Excel, CSV, XML, JSON and PDF. Power BI imports and caches the data in the .PBIX file using robust compression methods. Power BI adds a look to the process. Its drag and drop operation is simple, and it offers capabilities that let users duplicate all formatting between visualizations that are comparable. Stream analytics can be done in real time using Power BI. It enables users to access real-time analytics by collecting data from various sensors and social media platforms, ensuring that users are always prepared to make business decisions.

PART 2:

LISTS, DICTIONARIES AND LOOPING

The purpose of this Product Price Management System is to assist store owners in effectively managing the inventory. This system consists of features like adding products, updating product information, viewing all the products available and stopping the program.

The product name, price and quantity can be entered by users in features of adding products. Dictionaries are used to store the information where product name serves as key while product price and quantity serves as value. This part includes data validation to guarantee data integrity and gives the user straightforward feedback.

Users have the option to update the existing product by changing the value of price and quantity. It contains product existence checks, data validation and selective updates based on user input. This guarantees that inventory data is correct and up to date. This feature remains the same if no new value is entered.

In view all products feature, a list of every product along with its current price and quantity are displayed by the system. This feature gives a simple and informative summary of the current inventory, allowing the user to easily check the available stock.

Users can stop the program by key in “4” in the system input. The console will display the message “Exit the program.” if the condition is true. The break statement is put into action, where it will immediately terminate the while loop and stop the program. In simple terms, the “Stop the program” feature gives an organized and regulated way to quit the inventory system when the user has completed their job.

PART 3:

FUNCTION AND CLASSES

Function, class and inheritance are used in this scenario. This program was constructed by applying inheritance principles, with Cat as the parent class and Kitten as the child class. The Kitten class has additional attributes in the program and it uses the properties from the Cat class. By entering the details of the properties at each class, the program will generate 5 instances that show each characteristics of a Cat and Kitten.

A storage function in a separate file is referred to as a module. The code that will be imported into the program is included in a file ending .py. Storing a function in a separate file allows to hide the details of the program code and focus on its higher-level logic and allows reuse of functions in many different programs. Module.py is the name of the Python module that was created. This module contained the Cat and Kitten classes.

Classes combine functions and data into a single package. Breed and name are the two attributes that are present in the parent class (Cat) in this program. A function is a block of code that is designed to do a certain task and def is the term used to define a function. __init__ is a Python method that is called automatically when a new instance is created. The self parameter is required in the method definition as a reference point for the instance.

For every instance of Cat, the breed and name are initialized using __init__. Breed stores the breed of the Cat, such as Siamese, Siberian, Maine Coon, Exotic Shorthair and Scottish Fold. Name stores the name of the Cat such as Angel, Ellie, Gladis, Honey and Sheba. describe_cat used to print the information of breed and name of the Cat.

The reusability of the code is provided by inheritance. It enables us to add more functionality to a class without modifying it. The Kitten class inherits from the Cat class using the function of super(). Age is an additional attribute of the Kitten class.

The __init__ of the Cat class is called first by the Kitten class using the function of super().__init__(breed,name). This guarantees that the inherited properties are initialized correctly. Age stores the age of the Kitten. describe_kitten used to print the information about the age of the Kitten.

In this program, importing an entire module was done using the import module code. By importing it, Cat class can be accessed. Five variables were created, which are my_angel, my_ellie, my_gladis, my_honey and my_sheba. The result of the module.Cat call was assigned to the variables. The program then proceeded by creating 5 instances from the imported module. Following the creation of each instance, the breed and name of the Cat were printed using the describe_cat function.

The program proceeded with a child class, which is Kitten. The function creates five kitten instances with a module.Kitten, which are breed, name and age. Breed is a string representing the Kitten's breed, name is a string representing the Kitten's name and age is an integer representing the age of the Kitten in years. describe_cat code inherited from the Cat class and it will print the Kitten's breed and name. describe_kitten code used to print the information about the Kitten which refers to the age.

The feature of inheritance in Python is a powerful tool that encourages modularity, code reusability and a hierarchy of class relationships. Inheritance allows extending classes, overriding methods and transforming features from parent classes. Classes in Python include all the basic features of object oriented programming, where it allows multiple base classes, a derived class can override any methods of its base classes and the ability to call the base class within the same name.

Modules serve as storage that group variables, classes and related functions into reusable components. It enables code separation, promotes code utilization and aids in project scalability. In Python, an instance of a class also known as an object. Both data members and methods will be accessible to the call of a class object. Large Python applications require the creation of more modular programs and reusable code that can help in facilitating.

This Python program clearly demonstrates the notion of inheritance in object-oriented programming. The parent class (Cat) and the child class (Kitten) established a hierarchical connection. The Kitten class inherits methods and attributes from the Cat class, allowing for specialization and code reuse.